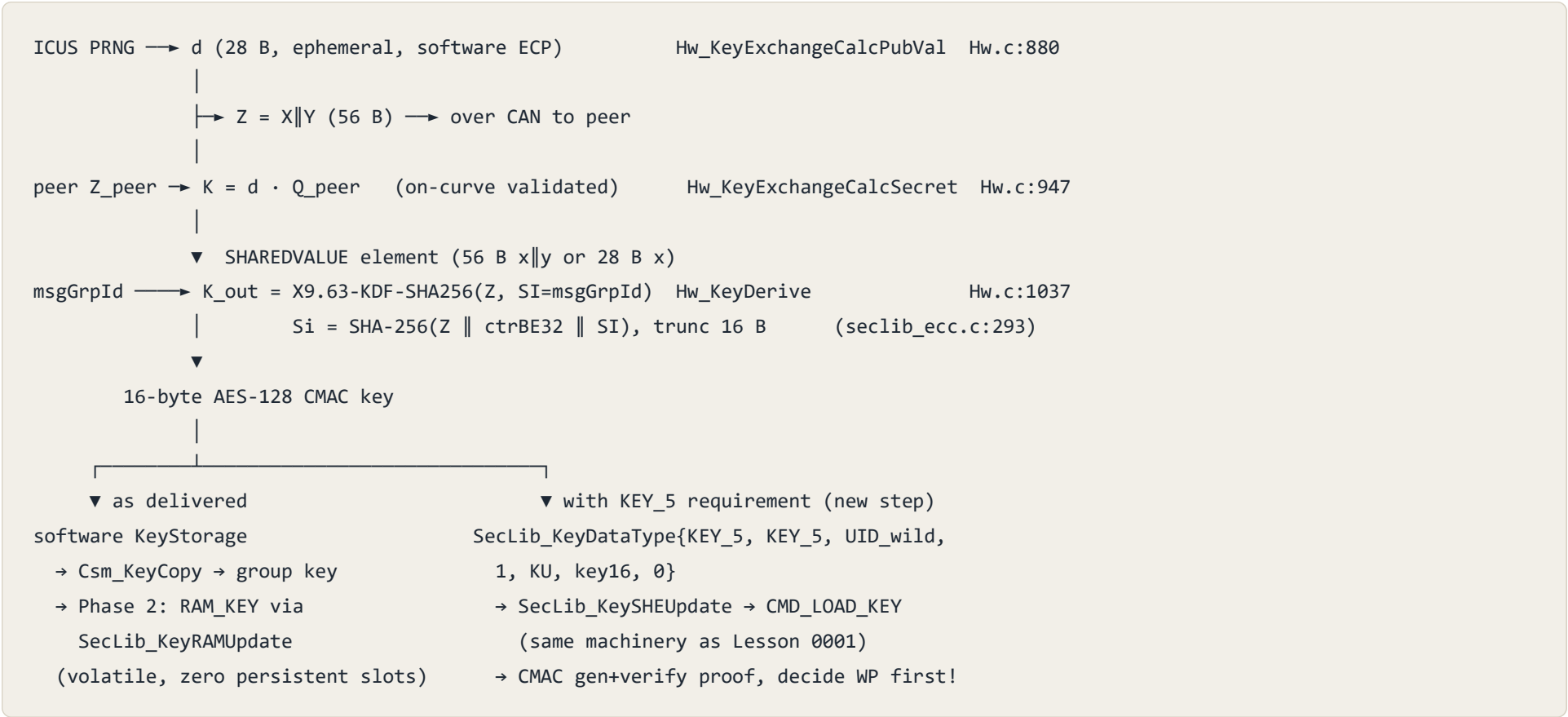


ECDH → ICUSE Slot Pipeline

Compressed essence of Lesson 0003. Grounded in `docs/ecc_integration_notes.md`.

1 • The chain, end to end



2 • KEY_5 parameter set

FIELD	VALUE	ANCHOR
Raw KEY_ID	0x08 (KEYMD = 0)	Cry_30_Rh850Icus_Types.h:75 · manual Table 4.1
keyId = authKeyId	KEY_5 (self-auth, empty slot ⇒ KEY_AuthID ≡ 0 ¹²⁸)	manual §4.4 p. 43 · secure_boot.c:439
Counter	1 first load; +1 on every renewal (monotonic, anti-replay)	manual §4.4
Flags	KU (CMAC-only) — WP only if the key never rotates	manual Table 4.2 · crypto_library_type.h:67–76
UID	wildcard (all-zero) while DWC = 0	crypto_library_type.h:60
Write path	SecLib_GenerateM1M2M3 + SecLib_KeySHEUpdate → CMD_LOAD_KEY	key_manager.c:99 · crypto_library.c:456 · Hw.c:1100

3 • Mandatory integration checklist

- Extend the SecLib slot whitelist:** add `case KEY_5:` to `SecLib_MACGenerate` (`crypto_library.c:238–246`) and `SecLib_MACVerify` (`crypto_library.c:319–327`) — otherwise every CMAC on KEY_5 fails with "keyId not supported".
- Bridge CryWrapper → security lib:** read the 16 derived bytes from the group-key element (or pass them directly) and fill `SecLib_KeyDataType`; wipe the software copy once `CMD_LOAD_KEY` succeeds.
- Respect the RAM_KEY ownership policy** while K1/K2 derivation bounces through the RAM slot (`key_manager.c:235, 287`) — no concurrent RAM_KEY user during the load.
- Decide WP before first provisioning** — one-way latch; rotating ECDH keys cannot live in a WP slot.
- Update the assurance workproduct:** notes §6 boundary ("persistent slots never written by the ECC path") is deliberately broken by this requirement — record the change and the justification.
- Verify:** CMAC generate + verify round-trip with the new slot (adapt `secure_boot.c:463–480`); then confirm SecOC frames authenticate.

4 • Assurance boundaries (from notes §7 — unchanged by KEY_5)

- ECC is **software** (mbedtls ECP 2.27.0 port + fork bignum); ICUSE has no ECC engine. "HPSE protection" claims must **not** be made for the negotiation path.
- Entropy source: ICUS PRNG via `SecLib_RandomGenerate` ; CS.00098 compliance to be confirmed by the security team.
- Hardening present: on-curve peer point validation (invalid-curve rejection), unbiased scalars, coordinate blinding, zeroization on every exit path.
- Ephemeral `d` lives in CPU RAM only, never in NVM or a persistent slot.

5 • Failure signals specific to this path

SIGNAL	CAUSE
"keyId not supported" (DLT 20019)	KEY_5 missing from the SecLib MAC switch — checklist item 1
E_VC_SECBOOT_ICUS_KEY_UPDATE_ERROR	M1M2M3 rejected: counter not greater than stored, CMAC mismatch, or UID/DWC violation
ERC_KEY_WRITE_PROTECTED (ERR5)	Attempt to renew a WP-flagged KEY_5 — the WP trap
SecLib_EccIsInitialized() false	SecLib_EccInit() not called after SecLib_Init() (notes §2)
Self-test KAT failure	Run SecLib_EccSelfTest() on bench after init — all 12 KATs must pass (notes §5)